Research Status Report

BLG vs. β-Casein Adsorption at the Air–Water Interface

IFSC2026 Poster Feasibility

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Purpose:	Decide whether to submit a poster to IFSC2026 (12–13 Nov 2026)
	Abstract deadline: 25 Sep 2026 | Poster: 60 cm(W) × 110 cm(H), portrait

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1. Executive Summary & Recommendation

Recommendation: submit the abstract. The 25 September deadline is not tight, and we have a poster-ready story today.

- **BLG is complete and poster-ready now.** The full 4.00 μs of unbiased atomistic MD is analysed, PBC-corrected, and internally validated. It stands on its own as a clean, publishable result with finished figures.
- **CAS (β-casein) is mid-pipeline and that is fine.** System preparation is two of five stages complete on a corrected topology; no production MD yet. A poster can present CAS honestly as *work in progress* — and the abstract itself needs no CAS results at all.
- **The abstract deadline drives nothing urgent.** An abstract describing the BLG finding plus the planned BLG/CAS comparison can be written from what already exists. There is more than two months of slack before 25 September.

The scientific headline is strong precisely because it is a *clean null result*: the mechanism we originally hypothesised was disproven cleanly, and we can state exactly where unbiased microsecond MD stops. That is a defensible, honest, and interesting story for a poster audience.

2. BLG Results — Complete and Validated

The 4.00 μs dataset. Four independent trajectories (one CENTER start + three replicas, each 1000 ns) totalling 4.00 μs of unbiased CHARMM36 MD of native BLG at a slab air–water interface. Across 8006 analysed frames we detect **613 protein–interface contact events** (97 from CENTER, 516 near-interface), of which only **6 are long-lived (≥10 ns)** — and every one of those six is *non-activated*: the protein touches the interface but does not begin to open, unfold, or commit.

2.1 The central result: contact without commitment

We originally hypothesised a *two-factor gate* — that surface-area opening (SASA) and a favourable orientation (θ) would fire together to trigger commitment to the interface. **This mechanism was disproven.** The apparent "gate opening" in the first-pass analysis was traced to a periodic-boundary (PBC) artifact: computing SASA without unwrapping split the protein across the box edge and inflated SASA to a spurious 45–62 nm^2. After correcting the unwrap step, the real SASA is **24–37 nm^2** (mean 28.95 nm^2), and there are **0 gate-open events across all 8006 frames**. The SASA–orientation correlation is statistically indistinguishable from zero: Pearson $r = +0.006$, block-bootstrap 95% CI **[−0.09, +0.11]** — ruling out any correlation with $|r| > 0.11$.

Why this matters. A near-zero correlation with a tight confidence interval is not a missing result — it is a positive statement: within microsecond unbiased sampling, native BLG presents a *pre-commitment contact ensemble* in which surface exposure and orientation are decoupled. The protein makes and breaks contact repeatedly without the two variables ever co-varying to drive commitment. The paper is therefore reframed as a precise *scope claim*: *"We characterise the pre-commitment contact* *ensemble of native BLG at the air–water interface and define precisely where unbiased microsecond MD stops."* This is a mechanistically clean, publication-ready null result — and it directly motivates the enhanced-sampling and comparative work that follows.

2.2 Supporting evidence — the protein stays folded and accessible

- **Globally compact throughout:** radius of gyration $R_g = 1.496 \pm 0.009\,\mathrm{nm}$ — essentially flat over $4\,\mu$s. BLG does not swell or partially unfold at the interface.
- **Hydrophobic patch does not open:** patch RMSD $\approx 0.24\,\mathrm{nm}$, flat — no progressive opening.
- **The calyx stays accessible:** the calyx-region SASA (the accessible hydrophobic pocket) is $3.81 \pm 0.46\,\mathrm{nm}^2$ ($n = 2001$, range 2.8–4.2 nm^2). The pocket neither collapses nor opens — it remains available the whole time. This is the "Accessible Calyx" half of the comparative story and the key metric CAS will be measured against.
- **Secondary structure preserved:** DSSP gives Helix 11.8% / Sheet 36.7% / Coil 51.5% — consistent with folded native BLG.
- **Physical sanity checks pass:** interfacial surface tension $51.9 \pm 38.5\,\mathrm{mN/m}$ (CENTER) matches the real TIP3P literature value (∼50–52 mN/m) almost exactly; the density profile shows bulk water at $1005\,\mathrm{kg/m^3}$ with the protein density peak ($72.5\,\mathrm{kg/m^3}$) sitting at $z = 1.76\,\mathrm{nm}$ near the interface.

Together these say the same thing four ways: over $4\,\mu$s, BLG contacts the interface frequently but remains folded, compact, and accessible — it never commits.

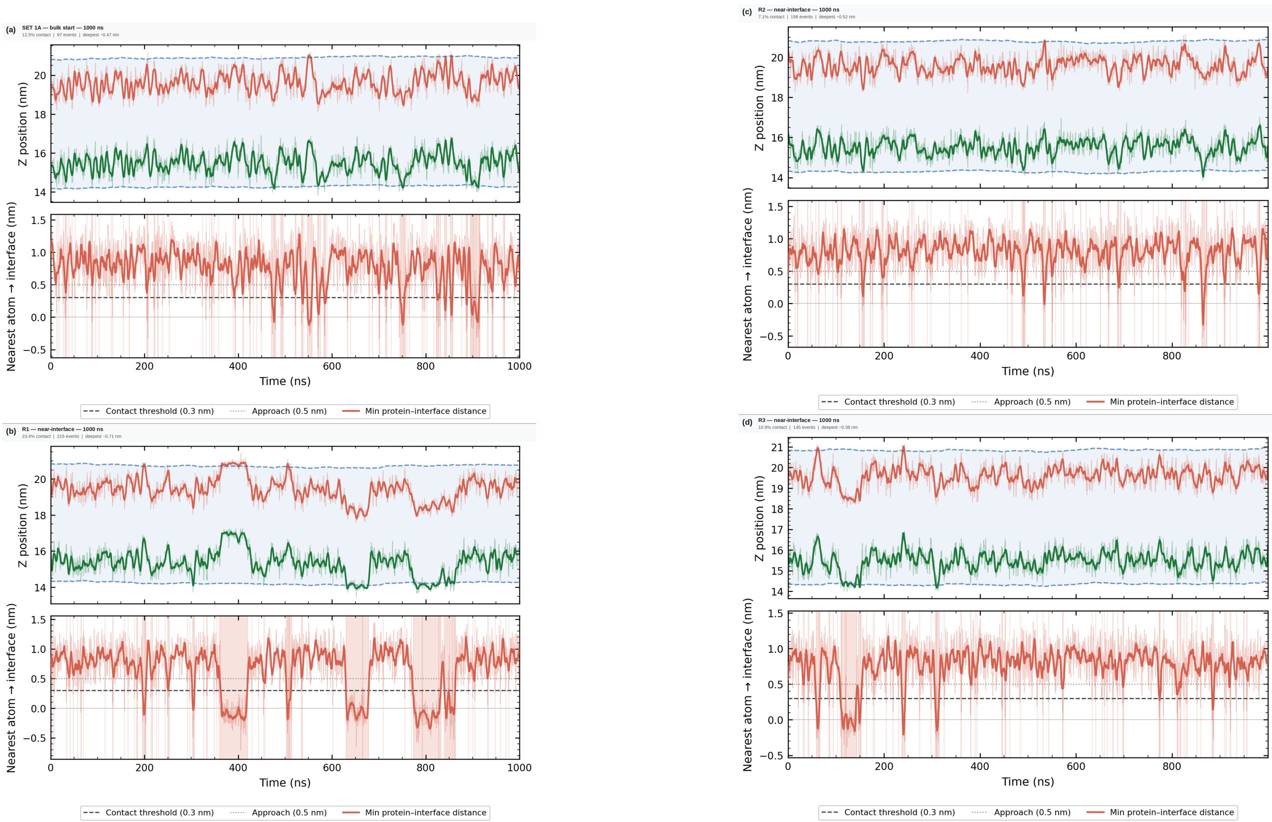

Figure 1: Contact detection across the $4\,\mu$s ensemble. Protein and interface z-positions (upper sub-panels) and the minimum protein–interface distance (lower sub-panels) for each trajectory. Shaded bands mark detected contact events (threshold 0.3 nm; approach 0.5 nm). Contact is frequent and transient across all runs — 613 events total, but only 6 last $\geq$10 ns and none progress to commitment.

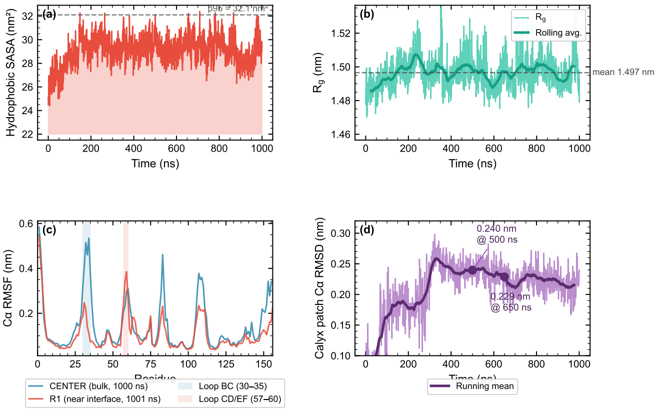

Figure 2: Activation / per-residue mobility. RMSF-based view of the contact ensemble. No sustained activation signature accompanies the long contact events — consistent with the non-activated, pre-commitment picture.

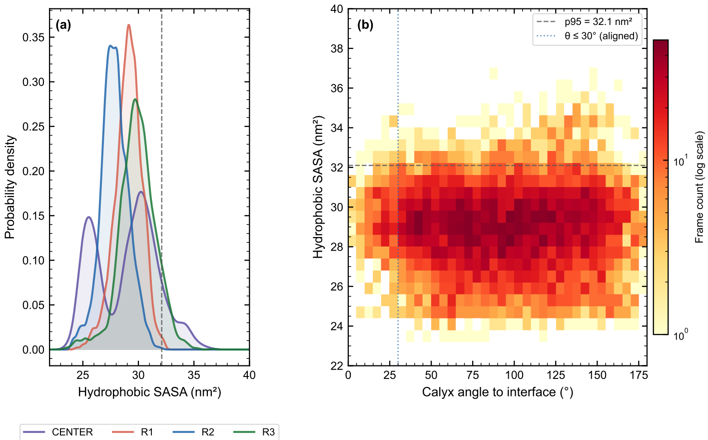

Figure 3: The null-correlation result (headline figure). Joint distribution of interfacial SASA and orientation angle θ across the PBC-corrected ensemble. The two variables are uncorrelated (Pearson $r = +0.006$, 95% CI $[-0.09, +0.11]$), ruling out the hypothesised two-factor gate. This decoupling of surface exposure and orientation is the quantitative core of the "contact without commitment" conclusion.

3. CAS (β-Casein) Status — Pipeline Progress

CAS is the second half of the planned comparative study. There are **no CAS science results yet** — the system is in preparation. Two of five preparation stages are complete on a corrected topology; production MD has not started.

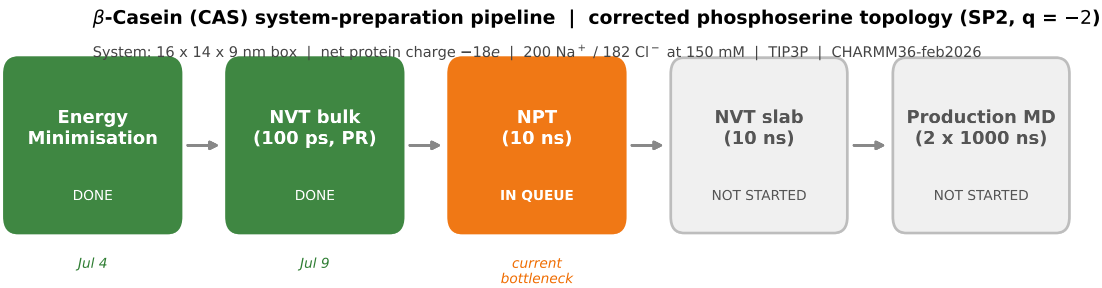

Figure 4: CAS system-preparation pipeline. Energy minimisation (4 Jul) and position-restrained NVT bulk equilibration (9 Jul, 100 ps, finished cleanly at $T = 297.9$ K with no NaN or LINCS warnings) are complete. NPT (10 ns) is the current bottleneck and needs a GPU/cluster submission; NVT slab and production MD (2 $\times$ 1000 ns, CENTER + R1) follow.

3.1 Why the pipeline restarted: a caught-and-fixed error (a sign of rigor)

CAS preparation was rebuilt from energy minimisation onward in June/July. This was **not lost time — it was a real scientific bug, caught and corrected.** β-casein's defining feature is its five phosphoserine residues (its key post-translational modification). These were initially built in the wrong protonation state — monoanionic ($q = -1$) — whereas at milk pH ($\sim$6.7) the phosphate group is dianionic ($q = -2$). We caught the error, verified the correct state against a literature-matched titration/pK_a argument, confirmed it with a clean energy-minimisation convergence, and rebuilt the topology (CHARMM36-feb2026 with an SP2 phosphoserine patch). Every downstream stage was then re-run on the corrected system. Catching this before production MD is exactly the discipline a reviewer would want to see — a wrong charge state on the residues that make β-casein interfacially interesting would have invalidated the whole comparison.

Current CAS system: 16 $\times$ 14 $\times$ 9 nm box, net protein charge $-18e$, neutralised with 200 Na^+ / 182 Cl^- at 150 mM NaCl, TIP3P water.

3.2 Planned headline metric and the comparative story

Once production MD exists, the headline CAS metric will be the SASA of the N-terminal (residues 1–25) amphipathic region — tracked the same way as BLG's calyx SASA — testing whether β-casein's disordered N-terminus stays "open" at the interface. This mirrors, and contrasts with, BLG's accessible-calyx result. The comparison title is already decided: *"An Accessible Calyx and an Open Chain: Contrasting Adsorption Strategies of β-Lactoglobulin and β-Casein at the Air–Water Interface."*

4. Timeline to 25 September — Calm and Confident

There is comfortable slack. Nothing below is on a critical path that endangers the abstract.

By	Milestone
Late July	Submit CAS NPT (10 ns) to GPU/cluster; then NVT slab. Choose poster framing with P.P.
August	CAS production MD (2×1000 ns) underway on cluster. BLG poster panels laid out from existing finalized figures (Figs. 2–4) — no new BLG analysis needed.
Early Sept	Draft abstract (BLG result + planned comparison). CAS presented as "in progress."
25 Sep	**Abstract deadline** — comfortably met.
Sep–Nov	Any CAS production data that lands gets added to the poster as it arrives; the poster does not depend on it being finished.
12–13 Nov	**IFSC2026 poster session.**

The BLG half of the poster could be built this week. The CAS half degrades gracefully: whatever production data exists by November strengthens it, and its absence does not weaken the abstract.

5. Open Questions for P.P.

1. **Poster framing.** BLG-only (a complete, clean story) vs. BLG + CAS-in-progress (the fuller comparative arc, with CAS shown as ongoing)? Both are defensible for a poster — which does P.P. prefer for an IFSC audience?
2. **Is "in progress" acceptable for CAS on a poster,** or would P.P. rather the poster wait until at least some CAS production data exists? (Recommendation: in-progress is standard and honest for a conference poster; the abstract needs no CAS results.)
3. **R2 replica handling** (Open Decision 1, expansion plan). R2's SASA trend is stable/downward while R1 and R3 trend upward. Keep all three and describe R2 as a non-committing pathway (most honest), or move R2 to supplementary? Note: all headline numbers were computed across CENTER + R1 + R2 + R3, so any change to R2's inclusion recomputes them.
4. **Applied-claim scope** (Open Decision 2, expansion plan). P.P. wanted an applied "how to improve adsorption" angle. The current data characterises the *pre-commitment* ensemble, so we can support a comparative design-principle statement (disorder + exposed hydrophobic patches lower the adsorption barrier) but not an explicit prescription. Is the comparative framing sufficient for the poster?
5. **Cluster resource** for CAS production MD (2×1000 ns): confirm which GPU/cluster allocation (ku-cluster vs. ku-ai) to target so NPT can be submitted.

All BLG numbers are PBC-corrected and validated (per project record; not to be changed without rerunning analysis). CAS status reflects file state as of 16 Jul 2026; no CAS production MD exists yet. Figures 2–4 are the finalized paper figures; the CAS pipeline figure was generated for this report.